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NovaCloud Vector Engine (NCVE) v4.2

Distributed Hybrid Search, HNSW Indexing & Real-Time Embedding Ingestion Pipeline

VERSION	RELEASE DATE	MAINTAINER	DEFAULT PORT
v4.2.0-stable	October 15, 2026	Core Storage Architecture Team	gRPC: 6334 / REST: 6333

1. System Overview & Core Architecture

The NovaCloud Vector Engine (NCVE) v4.2 is a cloud-native, distributed vector database designed for high-throughput Retrieval-Augmented Generation (RAG), semantic search, and anomaly detection workloads. NCVE combines dense vector similarity search via Hierarchical Navigable Small World (HNSW) graphs with sparse BM25 lexical scoring. Documents ingested through standard LangChain loaders are segmented into deterministic chunks, encoded into 1536-dimensional or 3072-dimensional floating-point vectors, and committed across raft-consensus shard groups with sub-millisecond write-ahead logging (WAL).

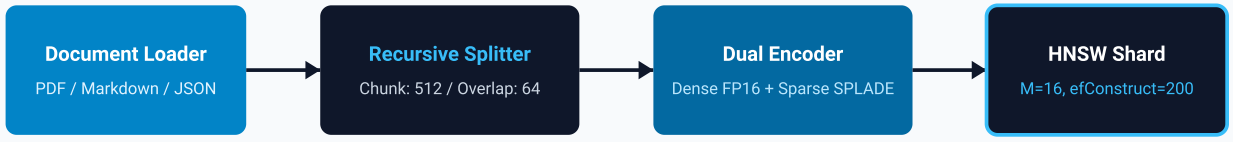


Figure 1.1: End-to-end document chunking, dual-vector encoding, and HNSW shard storage pipeline.

2. Indexing Parameters & Quantization

To optimize memory consumption across high-dimensional embeddings, NCVE v4.2 supports Scalar Quantization (SQ8) and Product Quantization (PQx64). Enabling SQ8 compresses 32-bit floating-point vectors (`float32`) into 8-bit signed integers (`int8`), reducing RAM overhead by 75% while maintaining 99.1% Recall@10 on standard MTEB retrieval benchmarks. Index construction utilizes three primary tuning hyperparameters: `m_links` (bidirectional links per node, default 16), `ef_construction` (candidate list size during build, default 200), and `ef_search` (dynamic candidate evaluation at query time, default 128).

3. REST & gRPC Endpoint Reference

All client requests must include the `X-NovaCloud-Api-Key` header authenticated via OAuth2 bearer tokens or static service principals. Payloads exceeding 16 MB are automatically rejected with status code `413 Payload Too Large`.

Endpoint	Method	Rate Limit	P99 Latency	Description & Required Payload Fields
`/v4/collections/create`	POST	50 req/min	42.0 ms	Initializes a new vector collection; requires `dimension`, `metric` (Cosine, Dot, L2).
`/v4/vectors/upsert`	PUT	2,500 req/min	14.5 ms	Batch inserts up to 1,000 vectors with JSON metadata payloads and page numbers.
`/v4/query/hybrid`	POST	5,000 req/min	18.2 ms	Executes Reciprocal Rank Fusion (RRF) across dense vector and BM25 lexical indices.
`/v4/vectors/delete`	DELETE	500 req/min	21.0 ms	Soft-deletes records by document UUID or metadata filter expression.
`/v4/cluster/health`	GET	10,000 req/min	3.1 ms	Returns Raft leader status, WAL lag, and active HNSW segment compaction tasks.

4. Hybrid Search Payload & Metadata Filtering

When querying NCVE from LangChain's `VectorStoreRetriever`, metadata filters are evaluated using a pre-filtering roaring bitmap index prior to graph traversal. Below is a canonical hybrid query payload combining dense semantic embeddings, sparse lexical tokens, and strict metadata filtering:

```
      { "collection_name": "enterprise_knowledge_base_prod", "query_vector": [-0.0142,
0.0891, -0.0432, 0.1124, "... (1536 dims)"], "sparse_indices": [104, 892, 4412, 9012],
"sparse_values": [0.42, 0.88, 0.71, 0.95], "alpha_fusion_weight": 0.75, "top_k": 5,
"filter": { "must": [ { "key": "department", "match": { "value": "Security_Engineering" } },
{ "key": "publication_year", "range": { "gte": 2025, "lte": 2026 } } ] } }
```

> **Operational Note for LangChain Text Splitters:** When testing `RecursiveCharacterTextSplitter` or `SemanticChunker` against this document, verify that Section 3's table rows stay associated with their header columns and that the JSON configuration block above is preserved as a single semantic unit without mid-key truncation.